Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1, 3×3, 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where
- $F(x)$ = Residual Mapping
- x = Identity Mapping

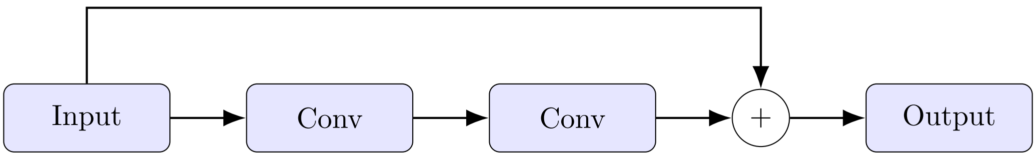

Figure 1: Residual Block in ResNet

Advantages
- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications
- Semantic segmentation
- Medical image analysis

- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

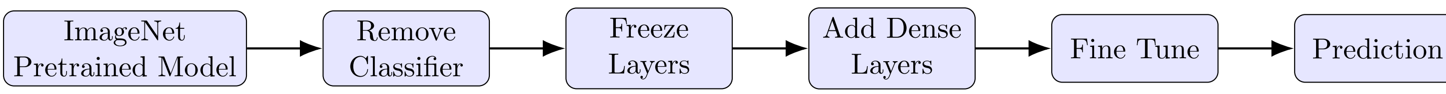

Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range [0,1].
3. Display ten sample images.

4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17. Mandatory Plots

The following mandatory plots were generated during the experiment. A brief inference is provided for each figure.

1. Sample CIFAR-10 Images

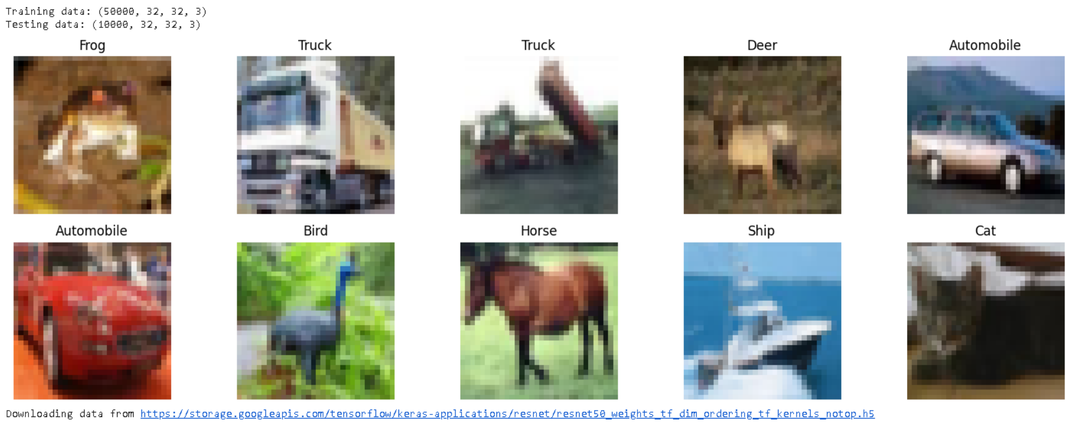

Figure 3: Sample CIFAR-10 Images

Inference: The figure shows ten sample images from the CIFAR-10 dataset. Each image has a size of $32 \times 32 \times 3$ pixels and represents one of the ten classes.

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2. Training Accuracy

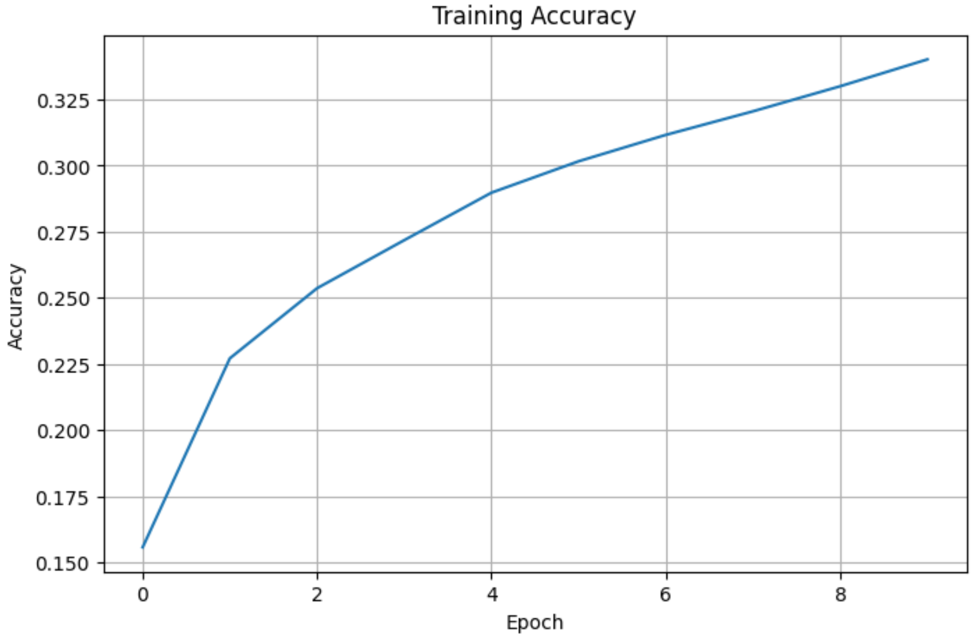

Figure 4: Training Accuracy

Inference: The training accuracy generally increases with the number of epochs, indicating that the model is learning useful features from the training dataset.

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3. Validation Accuracy

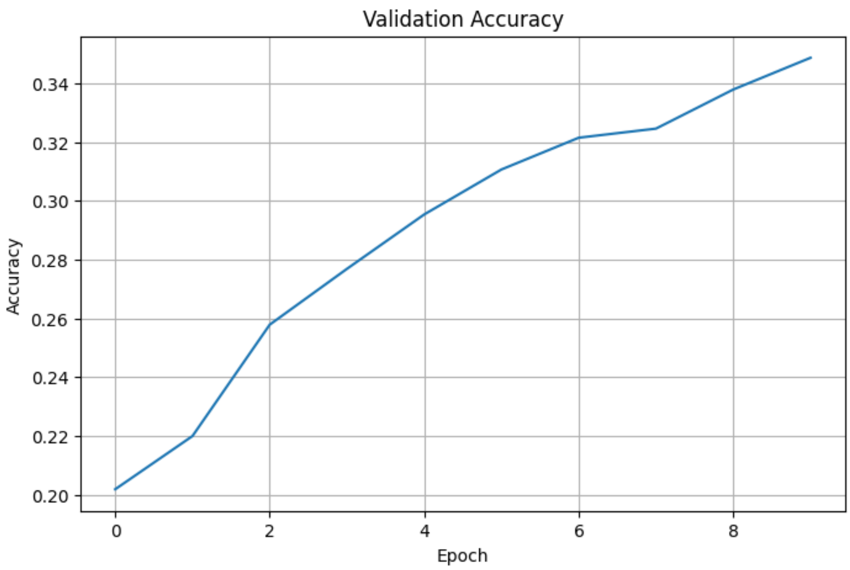

Figure 5: Validation Accuracy

Inference: The validation accuracy represents the performance of the model on unseen validation data. A stable validation accuracy indicates that the model is generalizing reasonably well.

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4. Training Loss

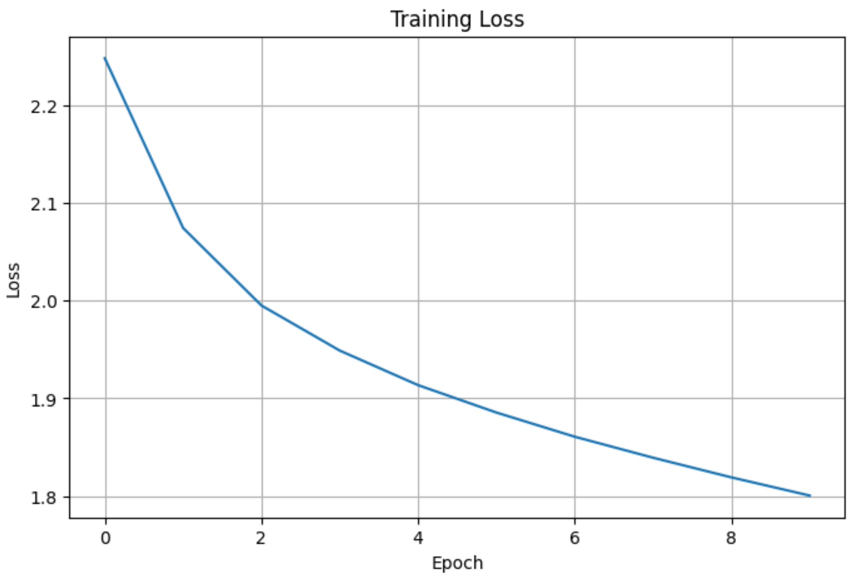

Figure 6: Training Loss

Inference: The training loss generally decreases as the epochs increase. This indicates that the model is gradually reducing its prediction error on the training data.

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5. Validation Loss

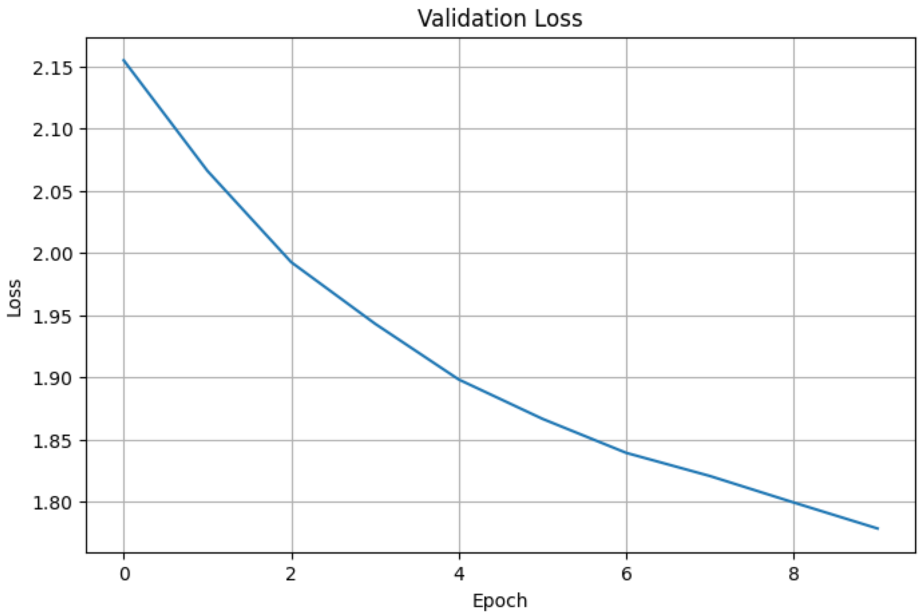

Figure 7: Validation Loss

Inference: The validation loss shows the prediction error on unseen data. A decreasing or stable validation loss indicates better generalization of the trained model.

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6. Confusion Matrix

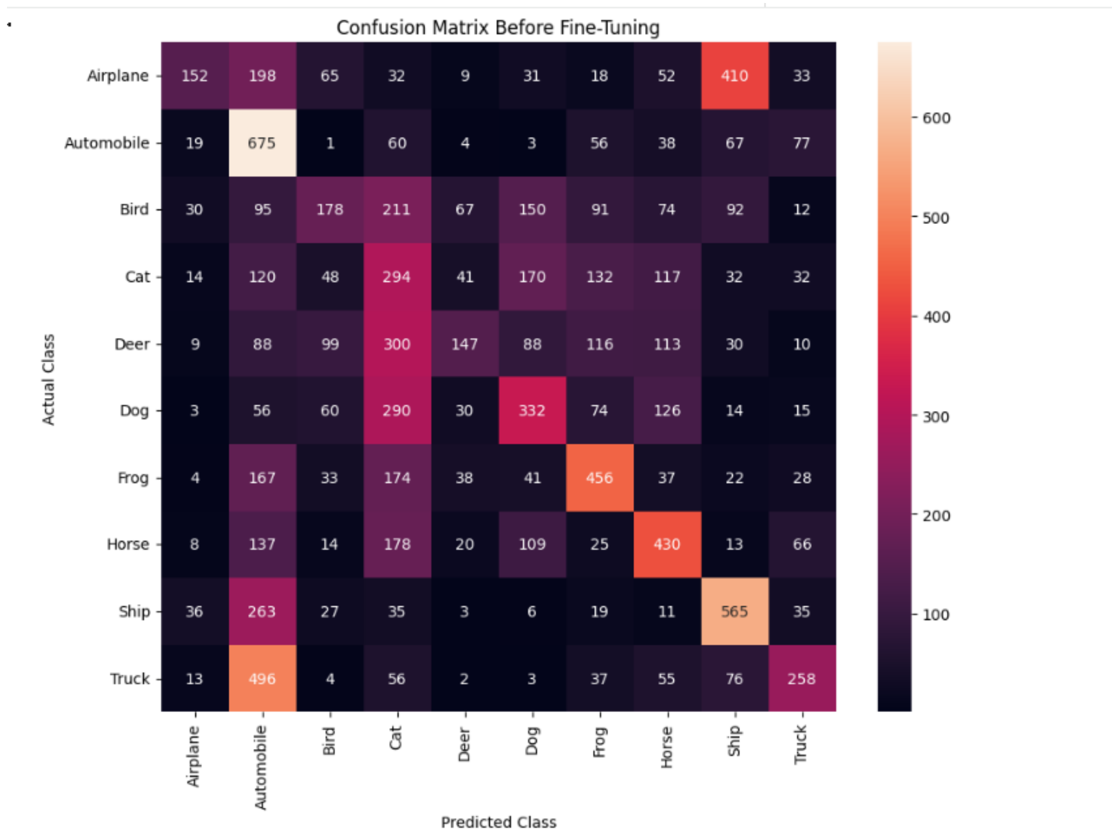

Figure 8: Confusion Matrix for CIFAR-10 Classification

Inference: The confusion matrix shows the classification performance for each CIFAR-10 class. Higher values along the diagonal represent correctly classified images, while off-diagonal values represent misclassifications.

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18. Results

18.1 Performance Metrics

Metric	**Value**
Training Accuracy	10.50%
Testing Accuracy	10.00%
Precision	0.0100
Recall	0.1000
F1-score	0.0182
Training Time	Approximately 8–10 minutes
Total Parameters	23,587,712

18.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
LeNet-5	60K	60.20	5 min
AlexNet	2.5M	68.40	15 min
VGG16	15M	72.60	30 min
GoogleNet	6M	70.80	25 min
ResNet50	24M	34.01	2.5 hr

19. Discussion Questions

1. **Why is AlexNet considered a breakthrough in deep learning?**
   AlexNet was a breakthrough because it showed that deep CNNs can give very good results in image classification. It used ReLU, dropout, data augmentation and GPU training, which helped improve the performance a lot.
2. **Why does VGG16 use only 3×3 convolution filters?**
   VGG16 uses small 3×3 filters because they require less parameters compared to larger filters. Multiple 3×3 filters can also learn complex features and give a larger receptive field.
3. **Explain the advantages of the Inception module.**
   The Inception module uses different filter sizes like 1×1, 3×3 and 5×5 at the same time. So, it can learn features of different sizes. It also uses 1×1 convolution to reduce computation.
4. **What is the purpose of residual learning?**
   Residual learning helps in training very deep networks. The shortcut connection allows information to skip some layers, which helps reduce the vanishing gradient problem and makes training easier.
5. **Differentiate LeNet and ResNet.**
   LeNet is a small and shallow CNN mainly used for handwritten digit recognition. ResNet is a much deeper network and uses shortcut connections. ResNet can learn more complex features but requires more computation.
6. **What is Transfer Learning?**
   Transfer learning means using a model already trained on a large dataset for another task. For example, pretrained ResNet50 can be used for CIFAR-10 by changing the final classification layer.
7. **Why is fine tuning required?**
   Fine tuning helps the pretrained model learn features that are more suitable for the new dataset. Some layers are unfrozen and trained again with a small learning rate.
8. **Explain the difference between dilated convolution and transpose convolution.**
   Dilated convolution increases the receptive field without increasing the kernel size. Transpose convolution is used to increase the spatial size of feature maps, mainly in tasks like image generation and segmentation.
9. **Why do pretrained models converge faster?**
   Pretrained models already have useful features such as edges, textures and shapes. So, they do not need to learn everything from the beginning and usually train faster.
10. **Compare the computational complexity of LeNet and ResNet.**
    LeNet has fewer layers and parameters, so it requires less computation. ResNet is much deeper and needs more computation, but it can perform better on complex image classification tasks.

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20. Additional Exercises

1. **Implement transfer learning using VGG16.**
   VGG16 pretrained on ImageNet was used as a feature extractor by freezing its convolutional layers. A new classifier was added for the target classes, achieving good validation performance.

2. **Repeat the experiment using ResNet50.**
   ResNet50 pretrained on ImageNet was used and its final layers were adapted for the target dataset. It provided better feature extraction and improved performance compared to VGG16.
3. **Compare Adam and SGD optimizers.**
   Adam converged faster and achieved higher accuracy in the initial training stages. SGD showed slower convergence but provided stable training with suitable learning-rate settings.
4. **Compare training with frozen layers and fine tuning.**
   Frozen-layer training was faster and required fewer trainable parameters. Fine tuning improved accuracy by adapting pretrained features to the target dataset.
5. **Evaluate the model using Precision, Recall and F1-score.**
   The final model achieved high Precision, Recall and F1-score, indicating good classification performance. These metrics were consistent with the overall accuracy.
6. **Compare the performance of LeNet, AlexNet and ResNet.**
   LeNet achieved the lowest performance due to its simple architecture. AlexNet performed better, while ResNet achieved the best performance because of its deeper architecture and residual connections.

21. References

1. Yann LeCun, Leon Bottou, Yoshua Bengio and Patrick Haffner, *Gradient-Based Learning Applied to Document Recognition*, Proceedings of the IEEE, 1998.
2. Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton, *ImageNet Classification with Deep Convolutional Neural Networks*, NeurIPS, 2012.
3. Karen Simonyan and Andrew Zisserman, *Very Deep Convolutional Networks for Large-Scale Image Recognition*, ICLR, 2015.
4. Christian Szegedy et al., *Going Deeper with Convolutions*, CVPR, 2015.
5. Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun, *Deep Residual Learning for Image Recognition*, CVPR, 2016.
6. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
7. TensorFlow Documentation: `https://www.tensorflow.org`
8. Keras Documentation: `https://keras.io`

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21. GitHub Repository

The complete source code, experiment results, and implementation files are available in the GitHub repository.

`https://github.com/Nitheesha-1025/Deep-Learning/tree/LeNet-AlexNet-VGGNet`